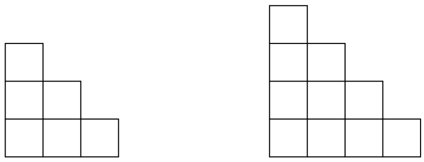
HK IMO Training 2018-2019 Problem Set 10

Suggested Solutions

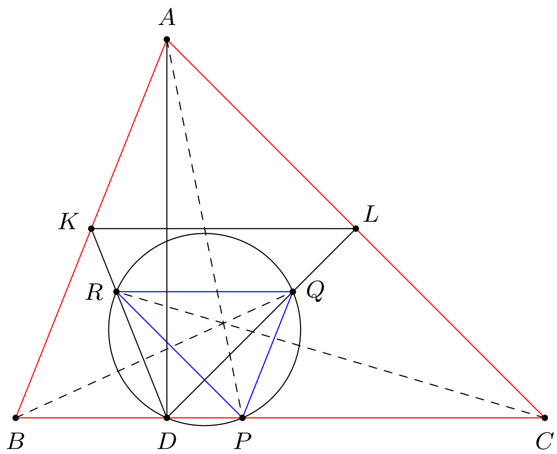
Problem 41. The two figures depicted below consisting of 6 and 10 unit squares, respectively, are called *staircases*.



Consider a 2018×2018 board consisting of 2018^2 cells, each being a unit square. Two arbitrary cells were removed from the same row of the board. Prove that the rest of the board cannot be cut (along the cell borders) into staircases (possibly rotated).

Solution. Enumerate the rows of the board with integers from 1 to 2018. We colour the cells of the board in horizontal strips of width 2 as follows: rows 1 and 2 are coloured red, rows 3 and 4 are coloured blue, rows 5 and 6 are coloured red, rows 7 and 8 are coloured blue, etc. If we disregarded the two cells removed from the board, both the number of red cells and the number of blue cells would be divisible by 4. Since the two cells are removed from the same row, they would have the same colour, hence after the removal we have that either the number of red cells is divisible by 4, while the number of blue cells is congruent to 2 modulo 4, or vice versa. In both cases, the numbers of red cells and of blue cells are not congruent modulo 4.
It now remains to observe that if a staircase, either of size 6 or 10, is placed on the board, then the difference of the numbers of red and blue cells covered by the staircase is always divisible by 4. This follows from a straightforward case study. Hence, if the board with the two cells removed could be tiled with staircases, then the difference of the numbers of red and blue cells would be divisible by 4, a contradiction. □

Problem 42. Let ABC be an acute-angled triangle with $AB < AC$, and let D be the foot of its altitude from A. Let R and Q be the centroids of the triangles ABD and ACD, respectively. Let P be a point on the line segment BC such that $P \neq D$ and the points P, Q, R and D are concyclic. Prove that the lines AP, BQ and CR are concurrent.



Solution 1. Without loss of generality, we may assume that P lies on the line segment CD. Let K, L be the midpoints of the sides AB, CA, respectively. As $\angle ADB = 90^\circ$ and K is the midpoint of AB, we have

$$\angle CBA = \angle KDB = \angle RQP. \qquad (*)$$

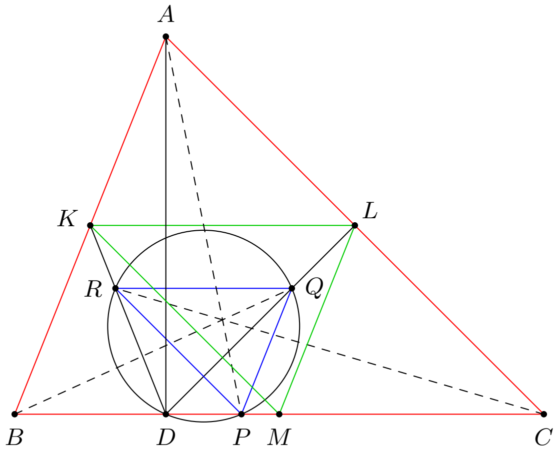
Similarly, we have

$$\angle ACB = \angle CDL = \angle PRQ. \qquad (**)$$

Furthermore, from the fact that

$$\frac{DR}{DK} = \frac{2}{3} = \frac{DQ}{DL}$$

we can see that $RQ \,/\!/\, KL$. Since $BC \,/\!/\, KL$ we also have $RQ \,/\!/\, BC$. This, together with angle equalities $(*)$ and $(**)$, implies that the sides PQ, QR, RP of $\triangle PQR$ are parallel to the sides AB, BC, CA of $\triangle ABC$, respectively. Obviously those triangles are not congruent, which means that there exists a homothety which maps $\triangle PQR$ to $\triangle ABC$. The centre of this homothety is the common intersection point of the lines AP, BQ, CR. □



Solution 2. The fact that triangles ABC, PQR are homothetic may be proven in a slightly different way, as follows. Let M be the midpoint of BC. Since the points A, D are symmetric with respect to the line KL, it follows that

$$\angle KML = \angle BAC = \angle KDL = \angle RPQ.$$

Consider a homothety h with center D and ratio $\dfrac{DK}{DR} = \dfrac{3}{2} = \dfrac{DL}{DQ}$. Then h maps the line segment QR to the line segment LK. Furthermore, h maps the point P to the point P' lying on the line BC such that $\angle LP'K = \angle QPR$. But there are exactly two points that satisfy those conditions, namely D and M. Since $P' \neq D$ we have $P' = M$ and therefore h maps $\triangle PQR$ to $\triangle MLK$. Composing h and the homothety centered at the centroid of ABC with ratio -2 which maps $\triangle MLK$ to $\triangle ABC$, we obtain a homothety with negative ratio which maps $\triangle PQR$ to $\triangle ABC$. □

Problem 43. Let n be a positive integer and let $a_1, \ldots, a_k$, where $k \geqslant 2$, be distinct integers in the set $\{1, \ldots, n\}$ such that n divides $a_i(a_{i+1} - 1)$ for $i = 1, \ldots, k-1$. Prove that n does not divide $a_k(a_1 - 1)$.

Alternative Formulation: Let G be a directed graph with n vertices $v_1, v_2, \ldots, v_n$, such that there is an edge going from v_a to v_b if and only if a and b are distinct and $a(b-1)$ is a multiple of n. Prove that this graph does not contain a directed cycle.

Solution 1. Suppose there is an edge from v_i to v_j. Then $i(j-1) = ij - i = kn$ for some integer k, which implies $i = ij - kn$. If $\gcd(i, n) = d$ and $\gcd(j, n) = e$, then e divides $ij - kn = i$ and thus e also divides d. Hence, if there is an edge from v_i to v_j, then $\gcd(j, n) \mid \gcd(i, n)$. If there is a cycle in G, say $v_{i_1} \to v_{i_2} \to \cdots \to v_{i_r} \to v_{i_1}$, then we have

$$\gcd(i_1, n) \mid \gcd(i_r, n) \mid \gcd(i_{r-1}, n) \mid \ldots \mid \gcd(i_2, n) \mid \gcd(i_1, n)$$

which implies that all these greatest common divisors must be equal, say be equal to t. Now we pick any of the i_k, without loss of generality let it be i_1. Then $i_r(i_1 - 1)$ is a multiple of n and hence also (by dividing by t), $i_1 - 1$ is a multiple of $\frac{n}{t}$. Since i_1 and $i_1 - 1$ are relatively prime, also t and $\frac{n}{t}$ are relatively prime. So, by the Chinese remainder theorem, the value of i_1 is uniquely determined modulo $n = t \cdot \frac{n}{t}$ by the value of t. But, as i_1 was chosen arbitrarily among the i_k, this implies that all the i_k have to be equal, a contradiction. □

Solution 2. If a, b, c are integers such that $ab - a$ and $bc - b$ are multiples of n, then also $ac - a = a(bc - b) + (ab - a) - (ab - a)c$ is a multiple of n. This implies that if there is an edge from v_a to v_b and an edge from v_b to v_c, then there also must be an edge from v_a to v_c. Therefore, if there are any cycles at all, the smallest cycle must have length 2. But suppose the vertices v_a and v_b form such a cycle, i. e., $ab - a$ and $ab - b$ are both multiples of n. Then $a - b$ is also a multiple of n, which can only happen if $a = b$, which is impossible. □

Solution 3. Suppose there was a cycle $v_{i_1} \to v_{i_2} \to \cdots \to v_{i_r} \to v_{i_1}$. Then $i_1(i_2 - 1)$ is a multiple of n, i. e., $i_1 \equiv i_1 i_2 \pmod{n}$. Continuing in this manner, we get $i_1 \equiv i_1 i_2 \equiv i_1 i_2 i_3 \equiv i_1 i_2 i_3 \cdot \ldots \cdot i_r \pmod{n}$. But the same holds for all i_k, i. e., $i_k \equiv i_1 i_2 i_3 \cdot \ldots \cdot i_r \pmod{n}$. Hence $i_1 \equiv i_2 \equiv \cdots \equiv i_r \pmod{n}$, which means $i_1 = i_2 = \cdots = i_r$, a contradiction. □

Problem 44. Determine all functions $f : \mathbb{R} \to \mathbb{R}$ such that the equality

$$f(\lfloor x \rfloor y) = f(x) \lfloor f(y) \rfloor \tag{1}$$

holds for all $x, y \in \mathbb{R}$. (Here $\lfloor z \rfloor$ denotes the greatest integer less than or equal to z.)

Solution 1. First, setting $x = 0$ in (1) we get

$$f(0) = f(0) \lfloor f(y) \rfloor \tag{2}$$

for all $y \in \mathbb{R}$. Now, two cases are possible.

CASE 1. Assume that $f(0) \neq 0$. Then from (2) we conclude that $\lfloor f(y) \rfloor = 1$ for all $y \in \mathbb{R}$. Therefore, equation (1) becomes $f(\lfloor x \rfloor y) = f(x)$, and substituting $y = 0$ we have $f(x) = f(0) = C \neq 0$. Finally, from $\lfloor f(y) \rfloor = 1 = \lfloor C \rfloor$ we obtain that $1 \leqslant C < 2$.

CASE 2. If $f(0) = 0$, we consider two subcases.

- Suppose that there exists $0 < \alpha < 1$ such that $f(\alpha) \neq 0$. Then setting $x = \alpha$ in (1) we obtain $0 = f(0) = f(\alpha) \lfloor f(y) \rfloor$ for all $y \in \mathbb{R}$. Hence, $\lfloor f(y) \rfloor = 0$ for all $y \in \mathbb{R}$. Finally, substituting $x = 1$ in (1) provides $f(y) = 0$ for all $y \in \mathbb{R}$, thus contradicting the condition $f(\alpha) \neq 0$.
- Conversely, we have $f(\alpha) = 0$ for all $0 \leqslant \alpha < 1$. Consider any real z; there exists an integer N such that $\alpha = \dfrac{z}{N} \in [0, 1)$ (one may set $N = \lfloor z \rfloor + 1$ if $z \geqslant 0$ and $N = \lfloor z \rfloor - 1$ otherwise). Now, from (1) we get $f(z) = f(\lfloor N \rfloor \alpha) = f(N) \lfloor f(\alpha) \rfloor = 0$ for all $z \in \mathbb{R}$.

Finally, a straightforward check shows that all the obtained functions satisfy (1). $\square$

Solution 2. Assume that $\lfloor f(y) \rfloor = 0$ for some y; then the substitution $x = 1$ provides $f(y) = f(1) \lfloor f(y) \rfloor = 0$. Hence, if $\lfloor f(y) \rfloor = 0$ for all y, then $f(y) = 0$ for all y. This function obviously satisfies the problem conditions. So we are left to consider the case when $\lfloor f(a) \rfloor \neq 0$ for some a. Then we have

$$f(\lfloor x \rfloor a) = f(x) \lfloor f(a) \rfloor, \quad \text{or} \quad f(x) = \frac{f(\lfloor x \rfloor a)}{\lfloor f(a) \rfloor}. \tag{3}$$

This means that $f(x_1) = f(x_2)$ whenever $\lfloor x_1 \rfloor = \lfloor x_2 \rfloor$, hence $f(x) = f(\lfloor x \rfloor)$, and we may assume that a is an integer. Now we have

$$f(a) = f\left(2a \cdot \frac{1}{2}\right) = f(2a) \left\lfloor f\left(\frac{1}{2}\right) \right\rfloor = f(2a) \lfloor f(0) \rfloor;$$

this implies $\lfloor f(0) \rfloor \neq 0$, so we may even assume that $a = 0$. Therefore equation (3) provides

$$f(x) = \frac{f(0)}{\lfloor f(0) \rfloor} = C \neq 0$$

for each x. Now, condition (1) becomes equivalent to the equation $C = C \lfloor C \rfloor$ which holds exactly when $\lfloor C \rfloor = 1$.

Finally, a straightforward check shows that all the obtained functions satisfy (1). $\square$